# Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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## PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

## CORE COMPETENCIES

Avionics (MIL-STD-1553 / ARINC 429, Avionics integration) · Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems) · Embedded systems (Embedded C, ESP32 / STM32 microcontrollers) · Edge AI (NVIDIA Jetson deployment) · Drones & UAV systems (UAV payload integration) · AI / machine learning (PyTorch, Deep learning) · IoT (MQTT and Node-RED, InfluxDB and Grafana) · Computer vision (Image super-resolution, Object detection) · GPU computing (CUDA / GPU computing) · Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation) · Software & tools (Python, Docker and Linux) · Technical leadership (Team leadership)

## PROFESSIONAL EXPERIENCE

**Scientist E** — Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

*Additional charge:* **Workshop In-charge**, Upper Air Division Workshop — Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

*Nodal officer:* **Nodal Officer, IoT and AI/ML Initiatives** — Jan 2021 – Present

**Senior Design Engineer** — Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

**Graduate Engineer** — Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

## KEY PROJECTS

**Edge super-resolution pipeline for UAV imagery** *(Ongoing)* — Sep 2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

**Technologies:** NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

## EDUCATION

**Ph.D., Electrical and Electronics Engineering** — 2021 – ongoing

Northfield Institute of Science and Technology

**Thesis:** Lightweight deep networks for real-time enhancement of aerial imagery

**M.Tech., Avionics** — 2010



*Fictional sample profile — not a real person*

Coastal Institute of Aeronautical Engineering, Chennai
**Grade:** CGPA 8.6/10
**Thesis:** Fault-tolerant data acquisition for airborne test instrumentation

**B.E., Electronics and Communication Engineering** 2004
Northfield Institute of Science and Technology, Northfield University, Pune
**Grade:** First Class with Distinction

## PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

## LEADERSHIP & RESPONSIBILITIES

**Supervisor, M.Tech. interns in applied machine learning** 2022 – Present
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

**Chair, Technical Evaluation Committee for radiosonde procurement** Apr 2023 – Feb 2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

## AWARDS & HONOURS

**Best Paper Award** 2023
Example Conference on Instrumentation (ECI 2023)

## CERTIFICATIONS & TRAINING

**Deep Learning for Edge Devices** Mar 2022
Example Academy of Applied Computing
**Duration:** 40 hours
**Credential:** EAAC-DL-0042
**Valid until:** Mar 2029

**Workshop on Efficient Deep Learning for Embedded Vision** Dec 2023
Example Conference on Instrumentation (ECI 2023)
**Duration:** 2 days

**National Seminar on Upper-Air Observing Systems** Aug 2021
Example Society of Atmospheric Instrumentation
**Duration:** 1 day

**Mid-Career Training Programme for Scientists** Feb 2020
Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences)
**Duration:** 10 days

## TECHNICAL SKILLS

**Avionics:** MIL-STD-1553 / ARINC 429, Avionics integration
**Scientific instrumentation:** Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
**Embedded systems:** Embedded C, ESP32 / STM32 microcontrollers
**Edge AI:** NVIDIA Jetson deployment
**Drones & UAV systems:** UAV payload integration
**AI / machine learning:** PyTorch, Deep learning
**IoT:** MQTT and Node-RED, InfluxDB and Grafana

## LANGUAGES

English, Hindi


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